

# Kisan Sethu — Plant Disease Classification Report

38-class evaluation report | Total test samples: 17,572

| Class                                      | Precision | Recall | F1-Score | Support |
|--------------------------------------------|-----------|--------|----------|---------|
| Apple___Apple_scab                         | 1.0000    | 1.0000 | 1.0000   | 504     |
| Apple___Black_rot                          | 1.0000    | 1.0000 | 1.0000   | 497     |
| Apple___Cedar_apple_rust                   | 1.0000    | 1.0000 | 1.0000   | 440     |
| Apple___healthy                            | 1.0000    | 1.0000 | 1.0000   | 502     |
| Blueberry___healthy                        | 1.0000    | 1.0000 | 1.0000   | 454     |
| Cherry___Powdery_mildew                    | 1.0000    | 1.0000 | 1.0000   | 421     |
| Cherry___healthy                           | 1.0000    | 1.0000 | 1.0000   | 456     |
| Corn___Cercospora_leaf_spot Gray_leaf_spot | 0.9975    | 0.9902 | 0.9939   | 410     |
| Corn___Common_rust                         | 0.9979    | 1.0000 | 0.9990   | 477     |
| Corn___Northern_Leaf_Blight                | 0.9916    | 0.9958 | 0.9937   | 477     |
| Corn___healthy                             | 1.0000    | 1.0000 | 1.0000   | 465     |
| Grape___Black_rot                          | 0.9958    | 1.0000 | 0.9979   | 472     |
| Grape___Esca_(Black_Measles)               | 1.0000    | 0.9958 | 0.9979   | 480     |
| Grape___Leaf_blight_(Isariopsis_Leaf_Spot) | 1.0000    | 1.0000 | 1.0000   | 430     |
| Grape___healthy                            | 1.0000    | 1.0000 | 1.0000   | 423     |
| Orange___Haunglongbing_(Citrus_greening)   | 1.0000    | 1.0000 | 1.0000   | 503     |
| Peach___Bacterial_spot                     | 1.0000    | 1.0000 | 1.0000   | 459     |
| Peach___healthy                            | 1.0000    | 1.0000 | 1.0000   | 432     |
| Pepper,_bell___Bacterial_spot              | 1.0000    | 1.0000 | 1.0000   | 478     |
| Pepper,_bell___healthy                     | 1.0000    | 1.0000 | 1.0000   | 497     |
| Potato___Early_blight                      | 1.0000    | 1.0000 | 1.0000   | 485     |
| Potato___Late_blight                       | 1.0000    | 1.0000 | 1.0000   | 485     |
| Potato___healthy                           | 1.0000    | 1.0000 | 1.0000   | 456     |
| Raspberry___healthy                        | 1.0000    | 1.0000 | 1.0000   | 445     |

| Class                                         | Precision     | Recall        | F1-Score      | Support       |
|-----------------------------------------------|---------------|---------------|---------------|---------------|
| Soybean___healthy                             | 1.0000        | 1.0000        | 1.0000        | 505           |
| Squash___Powdery_mildew                       | 1.0000        | 1.0000        | 1.0000        | 434           |
| Strawberry___Leaf_scorch                      | 1.0000        | 1.0000        | 1.0000        | 444           |
| Strawberry___healthy                          | 1.0000        | 1.0000        | 1.0000        | 456           |
| Tomato___Bacterial_spot                       | 0.9976        | 0.9929        | 0.9953        | 425           |
| Tomato___Early_blight                         | 1.0000        | 0.9958        | 0.9979        | 480           |
| Tomato___Late_blight                          | 0.9978        | 0.9978        | 0.9978        | 463           |
| Tomato___Leaf_Mold                            | 0.9979        | 1.0000        | 0.9989        | 470           |
| Tomato___Septoria_leaf_spot                   | 0.9977        | 0.9977        | 0.9977        | 436           |
| Tomato___Spider_mites Two-spotted_spider_mite | 0.9932        | 1.0000        | 0.9966        | 435           |
| Tomato___Target_Spot                          | 1.0000        | 0.9934        | 0.9967        | 457           |
| Tomato___Tomato_Yellow_Leaf_Curl_Virus        | 0.9939        | 1.0000        | 0.9969        | 490           |
| Tomato___Tomato_mosaic_virus                  | 1.0000        | 0.9978        | 0.9989        | 448           |
| Tomato___healthy                              | 0.9979        | 1.0000        | 0.9990        | 481           |
| <b>Accuracy</b>                               | <b>0.9989</b> | <b>0.9989</b> | <b>0.9989</b> | <b>17,572</b> |
| <b>Macro Average</b>                          | <b>0.9989</b> | <b>0.9989</b> | <b>0.9989</b> | <b>17,572</b> |
| <b>Weighted Average</b>                       | <b>0.9989</b> | <b>0.9989</b> | <b>0.9989</b> | <b>17,572</b> |

## Summary Metrics

| Metric                    | Value         |
|---------------------------|---------------|
| <b>Overall Accuracy</b>   | <b>99.89%</b> |
| <b>Macro Precision</b>    | <b>99.89%</b> |
| <b>Macro Recall</b>       | <b>99.89%</b> |
| <b>Macro F1-Score</b>     | <b>99.89%</b> |
| <b>Weighted Precision</b> | <b>99.89%</b> |

|                   |        |
|-------------------|--------|
| Weighted Recall   | 99.89% |
| Weighted F1-Score | 99.89% |
| Total Samples     | 17,572 |
| Number of Classes | 38     |